

Database Management Systems Lab (IT4PC_LR2)

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Problem statement : Hotel Booking System

The project is a dynamic booking platform designed to manage high-volume property data. It functions as a bridge between **Service Providers** (Hotels/Host Owners) and **Guests**, utilizing a relational database to ensure data accuracy and secure transactions.

● CRUD Operations Spotlight

The functionality of this system is best demonstrated through the four fundamental CRUD operations:

CREATE: Data Entry

- **Host Action:** When a property owner adds a new villa or hotel, the system performs an INSERT operation.
- **Constraint Check:** The DBMS validates that the Title and Country are present and that the listing_id does not already exist.

READ: Data Retrieval & Logic

- **Filtered Search:** Fetching records based on Location or Country.
- **Derived Attributes (GST Logic):** This project demonstrates Computed Columns or View Logic. When a user toggles the GST button, the system performs a SELECT operation that applies a mathematical formula:

Price after toggling : $\text{new price} = \text{original price} + (\text{original price}) * \text{GST}$

This avoids storing redundant data (Total Price) by calculating it dynamically from the stored Base Price.

UPDATE: Data Maintenance

- **Price Adjustment:** Hosts can perform an UPDATE operation on the Base_Price field.
- **Status Updates:** Changing the availability of a property from "Available" to "Booked" during a transaction.

DELETE: Referential Integrity

- **Cascade Constraints:** If a property listing is removed from the system, the DBMS can be configured with ON DELETE CASCADE. This automatically removes all associated Reviews linked via the Foreign Key, preventing "orphaned records" in the database.

Security & Transactional Roles

The database enforces **Access Control** through roles:

- **Authentication:** Ensuring the user exists in the Users table.
- **Authorization:** Restricting UPDATE and DELETE privileges on the Listings table to only the authorized "Host" or "Admin" roles, while "Guests" are restricted to READ (Listings) and CREATE (Reviews).

Technical Summary for DBMS Evaluation

- **Relational Model:** Implements strict PK/FK mapping.
- **Field Constraints:** Uses NOT NULL and VARCHAR limits to optimize storage and data quality.
- **Dynamic Logic:** Uses SQL-level arithmetic to handle tax calculations without bloating the physical storage.

1. Listing Architecture & Management

The system allows property owners to list various types of accommodations (hotels, villas, pools). Each entry is governed by strict data constraints to maintain a clean and searchable database:

- **Primary Identification:** Every listing is assigned a `listing_id` as a **Primary Key**. This is a unique integer that prevents any data overlap.
- **Data Integrity:** Fields like **Title** and **Country** are set to "No Null," meaning they are mandatory for every entry. These are limited to 50 characters to optimize database storage and UI consistency.
- **Geographic Detail:** A **Location** field captures specific neighborhood or city data, allowing for granular searching.

2. Dynamic Pricing & GST Logic

A standout feature of this platform is the **Real-time Price Calculation**.

- **Base vs. Gross:** The database stores a per-night base price.
- **Tax Integration:** On the user interface, there is a **GST Toggle**. When activated, the system applies a calculation logic to the base price to show the total cost inclusive of taxes. This demonstrates the system's ability to handle conditional data rendering and mathematical operations in real-time.

3. Relational Feedback System

The review feature is built on a **One-to-Many relational model**.

- **Review Linking:** Using a Foreign Key system, every review is "tagged" to a specific `listing_id`. This creates a permanent link, ensuring that user feedback is strictly mapped to the correct property and cannot be misattributed.
- **Qualitative & Quantitative Data:** The system captures a 1–5 star rating and the author's name (up to 50 characters), providing future guests with social proof of the property's quality.

4. System Security & Role Logic

The platform manages large amounts of user data through two critical layers:

- **Authentication:** Verifying user identities to allow bookings and review submissions.
- **Authorization:** Differentiating between roles. For example, only a **Host** has the authorization to "Add a Listing," while a **Guest** is authorized to "View and Review."

Technical Summary

- **Backend Logic:** Handles the mapping of reviews to listings and the GST calculations.
- **Database Design:** Focuses on relational integrity (Primary/Foreign Keys) and field constraints (Varchar limits and Non-Null attributes).
- **User Interface:** Designed for transparency, allowing users to switch between tax-exclusive and tax-inclusive pricing.